

Counterparts

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Comments welcome

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Chapter 1

A-theories and B-theories

1.1 Two key questions about time and modality

The central faultline in the philosophy of time is between the ‘A-theory of time’ and the ‘B-theory of time’. (This singularly unhelpful terminology is the fault of McTaggart 1908.) Since this book is devoted to working out a view whose classification either as an A-theory or as a B-theory is difficult, we will need to be careful in characterising what is at stake. But let us begin with some slogans and picture thinking, to help convey an intuitive sense for the two positions.¹

<i>The A-theory of time</i>	<i>The B-theory of time</i>
The present time is metaphysically special.	All times are “on a par”.
There are facts about which times are past, present and future, over and above the facts about which times are before which.	There are no such further facts. Pastness, presentness and futurity “reduce” to beforeness and afterness.
Time genuinely “flows” and “passes”. There is “objective becoming”.	We live in a “block universe”.
The facts themselves change.	Change is just a certain kind of pattern in the facts.

There is a lot of murkiness here. Rather than spending a long time parsing the slogans, I will proceed directly to a pair of questions in the vicinity which I think are clear enough to make progress with. Having got the questions on the table and made a case that they are intelligible and worth debating,

¹(Obviously I will need lots of citations here. . . .)

we will return in section 1.3 to the questions whether there is any important further doctrine in the vicinity of the A-theory that does not follow from answering both questions positively, and whether there is any important further doctrine in the vicinity of the B-theory that does not follow from answering both questions negatively.

The first question is

T1 Are there temporarily true propositions?

Call those who answer ‘yes’ to T1 ‘propositional temporalists’, and those who answer ‘no’, ‘propositional eternalists’.

Propositional temporalism is intended to gloss the last of our A-theoretic slogans, ‘The facts themselves change’. The inference from the slogan to propositional temporalism seems straightforward. Surely we can agree that the proposition that *P* is true when, and only when, the fact that *P* obtains; so if the fact that *P* is one of the facts that change—if, that is, the fact that *P* obtains only some of the time—then the proposition that *P* must be true only some of the time. The reverse inference, from propositional temporalism to the slogan, is obviously unproblematic if we take ‘fact’ to be just another word for ‘true proposition’. But to sustain the inference, we only need the much weaker thought that the true propositions which *do* correspond to facts jointly entail all the other true propositions. For in that case, any time the truth value of a proposition changed, there would have to be some change in the totality of obtaining facts.² And this thought should be acceptable even to proponents of “sparse” theories of facts, according to which the only facts are facts about the instantiation of certain “fundamental” properties and relations. For it seems to be part of the point of the idea of fundamental properties and relations that all truths are entailed by the totality of truths about which things there are and which fundamental properties and relations they instantiate. (?? below will revisit the question whether there is a coherent interpretation of the slogan that does not follow from propositional temporalism.)

²I am assuming here that if $p_1 \dots p_n$ entail p_{n+1} —i.e. that if it is metaphysically necessary that p_{n+1} is true if $p_1 \dots p_n$ are—then it is *always* the case that p_{n+1} is true if $p_1 \dots p_n$ are true. This seems obvious: what is metaphysically necessarily the case is always the case. But it is, surprisingly, controversial. In some combined logics for tense and modality, for example Kaplan’s “logic of demonstratives” (Kaplan 1989), ‘necessarily *P*’ does not entail ‘always *P*’. The considerations which might lead someone to deny this inference are discussed at length in Dorr and Goodman n.d.

T1 is not a question about language. But it is intimately associated with a debate in the philosophy of language, about the kind of semantic account appropriate for a certain large class of ordinary sentences. The sentences in question are the *temporally variable* sentences: those which could be used literally at two different times in such a way that one utterance involved someone saying something true, and the other involved saying something false—‘Obama is president’, for example. Given propositional eternalism, the semantic functioning of these sentences cannot be helpfully captured by choosing a *single* proposition to call their “semantic value” (in the relevant language). The reasons for this are well expressed by Frege, the archetypal propositional eternalist:

But are there not thoughts which are true today but false in six months time? The thought, for example, that the tree there is covered with green leaves, will surely be false in six months time. No, for it is not the same thought at all. The words ‘This tree is covered with green leaves’ are not sufficient by themselves for the utterance, the time of utterance is involved as well. Without the time-indication this gives we have no complete thought, i.e. no thought at all. Only a sentence supplemented by a time-indication and complete in every respect expresses a thought. But this, if it is true, is true not only today or tomorrow but timelessly. (Frege 1918, p.)

To give a non-crazy semantic account of sentences like ‘The tree is covered with green leaves’ and ‘Obama is president’, propositional eternalists must classify them as *context-sensitive*. Like ‘I am a philosopher’ and ‘This is a cup’, they are flexible tools whose meaning allows them to be used to express many different propositions. Some literal utterances of ‘Obama is president’ express eternally true propositions, while others express eternally false propositions; no single proposition is expressed by every utterance of the sentence. I will call the claim that all temporally variable sentences are context-sensitive in this sense “temporal contextualism”.

Note that temporal contextualists are free to give a range of different answers to the question which propositions are expressed by particular utterances of ‘Obama is president’. The most obvious candidates are propositions, concerning some particular instant *t* or interval *T*, to the effect that Obama is president at *t* or throughout *T*. However, this ‘date theory’ is certainly not the only option for the temporal contextualist. According to another time-honoured view, the ‘token-reflexive theory’, a particular utter-

ance u of 'Obama is president' will express the proposition that Obama is president during u .

Propositional eternalists, as we have seen, are pretty much required to be temporal contextualists. Propositional temporalists will not be persuaded by the argument given above: it would be natural for a propositional temporalist to hold that 'Obama is president' context-insensitively expresses a single proposition which is true exactly when Obama is president. Of course, propositional temporalists might have some other reason for positing context-sensitivity in this sentence; they might even hold the radical view that *all sentences whatsoever* are context-sensitive. But their belief in temporarily true propositions at least opens up the possibility of a very different account of the range of propositions available to be expressed by such sentences. This makes a broad range of linguistic considerations potentially relevant to T1.

Nevertheless, it would be a mistake to think of T1 as merely a question about the best way to do semantics. Unlike 'semantic value', 'proposition' is not just a piece of jargon which different semantic theories are free to use as they see fit. It is firmly anchored in ordinary discourse by certain rules which license us in treating a wide range of sentences involving 'proposition' as equivalent to certain other sentences not involving 'proposition': for example, 'The proposition that snow is white is true' is to be treated as equivalent to 'Snow is white'. It is these anchoring rules that make T1 a question of general interest to metaphysics, as opposed to a question *merely* about the particular ontological category of propositions. Section 1.2 will attempt to systematise the rules in question, and to explain how they can be acceptable to both propositional temporalists and propositional eternalists.

The second question by which I propose to capture the A-theory/B-theory dispute is T2:

T2 Is the unit class of the present instant much *more natural* than the unit classes of most other instants?

Call those who answer 'yes' to T2 *temporal elitists*, and those who answer 'no' *temporal egalitarians*. Temporal elitism is intended to capture A-theoretic slogans like 'The present is metaphysically special', while temporal egalitarianism captures B-theoretic slogans like 'All times are on a par'.

The expression 'natural class' in T2 is used in the sense of Lewis 1983a: it contrasts with 'gerrymandered class'. The idea is best introduced by means of examples. The class of all electrons is very natural. The class of all

electrons except for the ones in my left toe together with all donkeys is much less natural. A natural class groups things together in a non-arbitrary way: it carves nature at its joints. According to the temporal elitist, a classification of the instants of time into (i) the present instant; (ii) past instants (those that come before the present); and (iii) future instant (those that come after the present) carves along a natural joint. It is much more natural than the following classification: (a) the instant at which my alarm clock started going off yesterday morning; (b) instants earlier than that instant; (c) instants later than that instant.³

In using the language of naturalness, I am not presupposing any particular metaphysical account of what the naturalness of a class consists in. Nor am I presupposing anything about the importance of such facts to the project of metaphysics. *Prima facie*, claims about relative naturalness seem to be somewhat vague; it is not obvious that such vague language is especially useful for the project of developing a systematic metaphysical theory. Its main use, rather, is similar to that of the expression ‘metaphysically special’ for which it serves as a gloss: it is supposed to facilitate comparison of different metaphysical theories, by providing a theoretically neutral label for a structural feature that could be shared by such theories even when they are otherwise very different, and employ very different theoretical primitives. Expressions like ‘privileged’ and ‘distinguished’ often play a similar role, as when people characterise relativity theory as including the idea that no frame of reference is “privileged”. The idea is to reject a large family of theories which can be seen as agreeing that there is a privileged frame of reference, while saying different things about what its privilege consists in (e.g. being picked out by the points of absolute space, being the frame in which ether-particles are at rest, being a frame for which frame-relative simultaneity coincides with absolute simultaneity . . .). In view of the pervasive role of such expressions in physics, the idea that all claims of comparative naturalness are false or unintelligible is an extremely radical one.

Why have I set up T2 as a comparison between the present instant and *most* other instants? The point was to not to rule out the possibility that there are some other “distinguished” instants in the time-series which can be singled out in a non-arbitrary way. For example, if there is a first instant or a last instant, they are obviously quite special—their unit classes are quite natural—and likewise, if there is a unique instant that is exactly halfway

³Note that the notion of naturalness I am working with is one on which the *complement* of a very natural class is itself quite natural, since it carves along the same joint.

between the first instant and the last instant, its unit class is pretty natural too. Other instants may be special in virtue of what happens at them—e.g. the end of the era of cosmic inflation, or the beginning of the Rapture, or what have you. While many A-theorists will probably want to say that *being present* makes for *more* naturalness than any of the other features that uniquely distinguish particular instant, this strong claim is hard to argue for, and its denial is clearly far too weak to vindicate the B-theoretic slogan that “all times are on a par”. For our purposes the vague ‘most’ claim looks to be of about the right strength to divide those who think that the present is “privileged” from those who think that it is “just one time among many”.

It is tempting to get around this issue by introducing a more expressive language for talking about naturalness. We might, for example, allow ourselves to speak of classes as being natural in different *respects*; we could then characterise the A-theory/B-theory debate as turning (at least in part) on the question there there is a *respect* in which the unit class of the present instant is *uniquely* natural. But the new ideology of “respects” raises many questions that it would be nice to avoid; and in any case, the negative answer to this question does not do a good job of capturing the B-theoretic slogan that “all times are on a par”, since it can be accepted by an annoying character who accepts that there is a respect in which the unit class of the present instant is uniquely natural but only on the boring grounds that *every unit class whatsoever* is uniquely natural in its own distinctive respect.⁴

Another worry about T2 as a device for capturing the A-theory/B-theory issue is that some B-theorists will answer ‘yes’ if you happen to catch them at the right time. For example, if you email a B-theorist asking T2, a B-theorist who thinks that the Rapture will occur at exactly 8.32 pm this evening might type ‘yes’ in response and strive to press ‘send’ at exactly 8.32 pm, reasoning that this will be a way of communicating the truth that the unit class of

⁴Another strategy for getting around problems associated with “distinguished” instants other than the present is to talk about natural *properties* rather than natural *classes*. We could characterise A-theorists as thinking that *presentness* is a very natural property, and B-theorists as denying this. The problem with this is that there is disagreement within the “all times are on a par” camp about how to talk about *properties*. Some (***cite adverbialists***) think of properties as things that need to be supplemented with an object *and a time* before we have a complete proposition. Those who talk like this might well agree that the property of presentness is extremely natural: it has the structurally very distinctive feature that necessarily, it yields a truth whenever you supplement it with the same instant of time twice over, and a falsehood otherwise. Focusing on the naturalness of classes avoids this distracting issue, since no-one thinks that the question whether an object belongs to a class “only makes sense relative to a time”.

8.32 pm is much more natural than the unit classes of most other times. To avoid this worry, we might be tempted to modify T2 by adding a temporal modifier, as follows:

(T2*) Is it *always* the case that the unit class of whichever instant is present is much more natural than the unit classes of most other instants?

The disadvantage of this formulation is that some people who clearly belong on the B-theoretic side of our classification will be tempted to answer 'yes' for the wrong reasons. When a sentence occurs in the scope of a tense operator like 'always', there is strong interpretative pressure to find a way of understanding the sentence in such a way that the operator is not redundant. This pressure might prompt a B-theorist to reinterpret 'natural' as it occurs in (T2*) as the kind of predicate that can interact non-trivially with 'always'—in one common framework, that would mean positing a hidden time variable bound by 'always' as one of the arguments of the predicate. Various interpretations of 'natural relative to t ' will suggest themselves to the creative. Where x is some entity (perhaps utterly undistinguished), we could say that class A is natural relative to x iff there is some natural class of ordered pairs O such that A comprises all and only those objects y for which $\langle x, y \rangle$ belongs to O . For example, if x is a blob of mud with a specific hard-to-describe blobby shape, the class of all objects with exactly that shape will not be very natural *simpliciter*, but it will be very natural *relative to x* , since the class of ordered pairs whose two elements are exactly the same shape is very natural. Since the class of all ordered pairs that consist of the same object twice over is (intuitively) very natural, this definition of relative naturalness strongly suggests that relative to any object x , the unit class of x is very natural. Thus, assuming they agree that every instant is present at itself (and why wouldn't they?), even B-theorists will give a positive answer to (T2*) if they interpret it as 'Is every instant t such that the unit class of the instant that is present at t is more natural *relative to t* than the unit classes of most other instants?' For this reason I think we will do better to focus on T2, which does not cry out in the same way for an interpretation in terms of a notion of naturalness relative to t . For our present purposes, it is more important to minimise the opportunities for misunderstanding than to come up with a formulation of the question that works even in the presence of bizarre hypotheses according to which some of the times at which we are having this discussion play some *permanently* distinguished role within the time series.

Questions T1 and T2 only succeed as articulations of basic questions about the metaphysics of time if we hold fixed certain background ontological assumptions, about *propositions* in the case of T1, and *instants of time* in the case of T2. When people reject these assumptions, their answers to T1 and T2 may reflect only their heterodox views about propositions and instants, rather than anything that is intuitively relevant to their categorisation as A-theorists or B-theorists. For example, nominalists, who deny the existence of propositions and/or instants of time on the grounds that there are no abstract entities of any sort, could still be recognisable either as A-theorists or as B-theorists, even though they will be unable to use T1 and T2 to articulate these views (unless they have some story about why it is sometimes legitimate to play along with ordinary ways of talking about abstract entities). This sort of situation is very common in philosophy. It is tempting to react to it by looking for perfectly neutral formulations of the relevant questions, whose answers will depend only on the subject matter we are concerned with without the need for any background assumptions. But since such neutrality is typically unachievable, the best way to pursue clarity is not to dispense with the need for any background assumptions, but to articulate the required assumptions as explicitly as possible. Section 1.2 below will attempt to do this for the background theory of propositions presupposed by T1, while ?? will attempt to do the same for the background theory of instants presupposed by T2. Once the required assumptions are on the table, we will at least be able to tell whether a given theorist rejects the assumptions, and will hopefully be better equipped to tackle the delicate enterprise of figuring out how to rephrase our questions in a such a way as to capture the underlying metaphysical issues in a form accessible to the theorist in question. If we are lucky, we will be able to find some systematic replacements for the distinctive vocabulary that occurs in our background theory of propositions ('proposition' and 'true'), or instants ('instant', 'past', 'present', 'future', 'before') such that the background theories come out true once the replacements are made, according to the theorist. In such cases we will likely find that the best way to classify our theorist is simply to look at their answers to the questions that result from making the relevant replacements in T1 and T2.⁵

⁵It is worth noting that temporal elitists and egalitarians are likely to disagree about the naturalness of many classes other than classes of instants. For example, in a world where objects often change their shape, it would be plausible for a temporal elitist to hold that the class of all *currently* spherical objects is much more natural than most other classes that have

Since we have two yes-no questions, there are in principle four combinations of views we need to think about. Two of the combinations—propositional temporalism plus temporal elitism, and propositional eternalism plus temporal egalitarianism—have a familiar feel; they are part of the stereotypical A-theoretic and B-theoretic packages. But must we also take seriously the hybrid views which answer ‘yes’ to one question and ‘no’ to the other? Although none of the central claims in the book will depend on this, I think there are strong arguments that these hybrids are non-viable.

The case against the combination of propositional eternalism and temporal elitism is fairly straightforward. Clearly, the temporal elitist will think that one can speak truly by uttering ‘The unit class of the present instant is more natural than those of most instants’ no matter when it is. So, for example, utterance ‘{2015} is more natural than {2014}’ will express truths in 2015 and falsehoods in 2014. As we have seen, propositional eternalists have to account for this kind of temporal variability in language by positing context-sensitivity. In the present case, the relevant source of context-sensitivity would be ‘more natural than’. But treating ‘more natural than’ as context-sensitive in *this* way would completely defeat the intended purpose of the predicate. It’s just as if a believer in Minkowski spacetime were to understand ‘privileged frame of reference’ as context-sensitive in such a way that anyone who says ‘There is exactly one privileged frame of reference, and right now I am at rest relative to it’ thereby speaks the truth. (The same point applies to ‘metaphysically special’, ‘distinguished’, etc.) It is central to the communicative role of these expressions that we avoid interpreting them as having this kind of extreme context-sensitivity; there is no point in using the expressions at all if our interlocutors are unwilling to exercise this minimal level of charity.

The combination of propositional temporalists and temporal egalitarianism has been much more popular, and my argument against it will be more controversial. It goes as follows. Say that an instant is ‘accurate’ iff the propositions that are true *at it* are all and only the *true* propositions. If propositional eternalism is true, all instants are accurate, since a proposition that is true at one instant is true at every instant. By contrast, if propositional temporalism is true, most instants are inaccurate. The sun is either shining or

in the past, or will in the future, contain all and only the spherical objects. By focusing on disagreements like this one, we could avoid the need for any assumptions about the ontology of instants. But it is not clear how, without mentioning instants, we could abstract away from the details of this example to a general thesis.

not. If it is, the proposition that the sun is shining is true, and all the instants at which it is false are inaccurate; if it is not, the proposition that the sun is not shining is true, and all the instants at which *it* is false are inaccurate. Among the instants that are accurate as regards whether the sun is shining, many are inaccurate as regards whether I am sitting. Clearly, at most a few instants are accurate in every respect: in fact, there is a strong case that the present instant is the *unique* accurate instant. On its face, accuracy seems like a very interesting feature for an instant to have. A classification that puts the one and only accurate instant in a category by itself seems to do a much better job of carving at the joints than one that, say, puts the instant that will be accurate 17 years from now into a category by itself. Thus accuracy would seem to be a rather natural property, and the class of all accurate instants would seem to be a rather natural class.

Given the popularity of the combination of propositional temporalism with temporal egalitarianism, this line of argument deserves some scrutiny. ?? will develop it more carefully, while section ?? will consider ways in which it might be resisted.

Our questions about the metaphysics of time have modal analogues:

M1 Are there contingently true propositions?

M2 Is the unit class of the actual world much more natural than the unit classes of most possible worlds?

Call those who answer 'yes' to M1 *propositional contingentists*, those who answer 'no' *propositional necessitarians*; those who answer 'yes' to M2 *modal elitists*, and those who answer 'no' to M2 *modal egalitarians*. I will also sometimes call those who answer 'yes' to both questions "modal A-theorists", and those who answer 'no' to both, "modal B-theorists".

One kind of propositional necessitarian is the Spinozist, who denies that we can ever speak truly by saying something of the form 'It is contingent whether ϕ ', or 'Not necessarily ϕ , and not necessarily not- ϕ '. But I will mostly be thinking about a different sort of propositional necessitarian, whose view is more closely modelled on the standard B-theoretic package in the philosophy of time. Just as standard propositional eternalists hold that 'It is only sometimes the case that ϕ ' can be true without there needing to be any proposition that is only sometimes true, these propositional necessitarians hold that 'It is contingent whether ϕ ' can be true without there needing

to be any proposition that is contingently true. Not being Spinozists, they agree that there are many possible worlds. Just as propositional eternalists must posit context-sensitivity in a sentence ϕ whenever there are two different times at which ϕ is uttered literally, at only one of which the person uttering ϕ thereby asserts a truth, so propositional necessitarians must posit context-sensitivity in ϕ whenever there are two different *possible worlds* at both which ϕ is uttered literally, such that only one of them is a world at which the person uttering ϕ thereby asserts a truth. For example, a propositional necessitarian will take the sentence 'Jupiter and Saturn never collide' to be context-sensitive: it is capable of being used, literally, to express both (necessary) truths and (necessary) falsehoods. The most straightforward account of this context-sensitivity would be as follows: at any given possible world w , people who utter 'Jupiter and Saturn never collide' thereby express the proposition that Jupiter and Saturn never collide *at w* : a proposition that, according to the standard way of talking about possible worlds, is necessary if true, and impossible if false.

(Some philosophers that propositions are "individuated by their truth conditions", i.e. that whenever it is necessary that proposition p is true iff proposition q is true, p is identical to q . The combination of this doctrine with propositional necessitarianism entails that there are only two propositions: one necessarily true, the other necessarily not true. This is quite bizarre, and hard to reconcile with the idea that propositions are "objects of the attitudes", an idea which will play an important part in the proposition role as we will lay it out in section 1.2. Thus, propositional necessitarians will probably want to adopt some more "fine-grained" account of the sufficient conditions for identity among propositions. This could, for example, be a Russellian conception of propositions as complexes with a structure parallel in some respects to that of the sentences that express them, or a Fregean conception on which propositions somehow build in the criteria people need to satisfy in order to bear propositional attitudes towards them. At any rate, it would be a mistake to assume that propositional necessitarianism has to go along with the view that there are only two propositions.)

The arguments sketched above that T1 and T2 should be answered in the same way have parallels for M1 and M2. From propositional contingentism to modal elitism: given propositional contingentism, it seems like we should be able to pick out the actual world as the unique *accurate* world; and this distinction seems quite an interesting one, metaphysically speaking; just the sort of thing that should make for a "joint in nature" between the actual

world and all the inaccurate world. From modal elitism to propositional contingentism: if we consider only its non-contingent characteristics, the actual world is not all that special; any facts about it that could single it out as special would have to be contingent ones. As in the temporal case, the latter inference has to deal with interference arising from other kinds of natural distinctions that could, in principle, be imputed to the actual world even by a propositional necessitarians. For example, the propositional necessitarian might hold, at a certain world w , that w necessarily enjoys the supreme distinction of being the best of all possible worlds. At every world, the wise realise that w is best, and that its unit class is quite natural. At w , uses of 'the actual world' refer to w , and those who are wise at w know this at w . So at w , the wise answer 'yes' to M2, despite being propositional necessitarians. This corresponds to the worry that certain propositional eternalists might answer 'yes' to T2 if you happen to ask them at an instant they regard as very special, e.g. at the instant at which they take the Rapture to begin. However, the issue is harder to dismiss in the modal case, since we can't get around the problem just by asking M2 again. Still, I think we will do better to tolerate this potential to misclassify those with eccentric views rather than attempting to elaborate the wording of M2 to control for the issue: the elaborations are apt to do more harm than good, by presenting new interpretative pitfalls even for those with non-eccentric views.

Despite the formal analogies between T1 and M1, and between T2 and M2, there is (at least) a salient sociological difference. In the temporal case, the B-theory is the dominant view, but A-theorists are a vocal minority. In the modal case, by contrast, almost everyone who has considered the question has come down on the side of propositional contingentism and modal elitism. (And there is no point in my making a secret of the fact that I am entirely in agreement with this near-consensus.) There is thus much to be learnt by considering the two debates in parallel, calibrating our assessment of the weight of any given consideration by comparison with its analogue in the other debate.

The best-known dissenter from the consensus in favour of modal elitism is Lewis (1986b). Lewis famously denies that there is any important metaphysical distinction between the thing he calls 'the actual world' and the other things he calls 'possible worlds'. However, it would be a mistake to take Lewis as our paradigm of a "modal B-theorist". For one thing, he endorses propositional contingentism. For another, as we will see in ??, the role played in Lewis's theory by the things he calls 'possible worlds' does not

perfectly fit the standard “possible world role”. So we must be on the alert lest we mistake some verbal difference regarding the use of this particular piece of philosophical jargon for the substantive dispute we are trying to zero in on.

The rest of this chapter will be organised as follows. First, section 1.2 will explain the background theory of propositions assumed by T1 and M1, and explain why it is not enough on its own to settle these questions. ?? will digress into the philosophy of language, carefully laying out the argument sketched above that propositional eternalists and necessitarians are committed to regarding certain kinds of sentences as context-sensitive. ?? will then turn to T2 and M2, laying out the background assumptions about instants of time and possible worlds which these questions assume. ?? will revisit the argument given above for thinking that these questions should be answered in the same way as T1 and M1, respectively. ?? and ?? will then consider attempts to make room for views which answer ‘yes’ to T1 but ‘no’ to T2, or ‘yes’ to M1 but ‘no’ to M2.

1.2 Propositions

To make sense of T1 and M1, and to explain why they are relevant not just to the investigation of the particular ontological category of *propositions* but to our larger conception of the phenomena of time and modality, we need to get on the table a certain role which all sides can take it for granted that propositions play.

The first and most central part of this role is the *propositional truth schema*:

Truth schema: The proposition that ϕ is such that for it to be true is for it to be the case that ϕ

This is a schema: a type whose instances are sentences. Its instances are sentences like ‘The proposition that snow is white is such that for it to be true is for it to be the case that snow is white’. Following standard practice, I will take the class of instances of any schema to be closed under necessitation and universal quantification. Thus, the following will also count as an instance of the Truth Schema: ‘Necessarily, for every x , the proposition that x is happy is such that for it to be true is for x to be happy’.

The status claimed on behalf of the sentences that are instances of the propositional truth schema is this: they must express truths, provided that any

ambiguity or context-sensitivity in the sentence substituted for the schematic letter ' ϕ ' is resolved uniformly in both occurrences. This restriction to uniform interpretations is one we standardly take for granted in setting forth schematic logical principles. For example, someone might in a favourable circumstance use the sentence 'Bill Bradley is tall and Bill Bradley is not tall' to say that Bill Bradley is tall for a politician but not tall for a basketball player (which is true). Similarly, under certain circumstances one might use 'If Bill Bradley is tall, he is tall' to say that if Bill Bradley is tall for a politician he is tall for a basketball player (which is false). Recognising this possibility is consistent with endorsement of the schema 'If ϕ then ϕ ' as part of the logic of 'if'. Similarly, we can accept the propositional truth schema while recognising that one could speak truly by uttering 'Bill Bradley is tall, and the proposition that Bill Bradley is tall is not true', or speak falsely by uttering 'If Bill Bradley is tall, the proposition that Bill Bradley is tall is true', provided that one understood the two occurrences of 'tall' non-uniformly.

The other part of the proposition role that I will be taking for granted is sometimes put by saying that propositions are 'objects of belief and assertion'. It turns out that people use 'object' in different ways, but the role I have in mind can be captured by the following schemas:

Belief Schema: The proposition that ϕ is such that to believe it is to believe that ϕ .

Assertion Schema: The proposition that ϕ is such that to assert it is to assert that ϕ .

The general rules given above for interpreting schemas apply here too. So, for example, the belief schema commits us to the claim that necessarily, for every x , the proposition that x is happy is such that to believe it is to believe that x is happy.

'Belief' and 'assert' belong to the category of *attitude verbs*, a category that also includes 'know', 'fear', 'hope', and 'wish'. But for most of these other verbs, the direct analogues of the Belief Schema and the Assertion Schemas cannot be endorsed. For some of the verbs, the problem is that the instances of the schema would be ungrammatical: 'I hope the proposition that snow is white' is ill-formed. For others, the problem is that the instances would be false: there may be some sophisticates who fear the proposition that the Earth is warming (they are "proposition-phobes"), but not everyone who fears that the Earth is warming is like this. I do not take this to count against

the claim that hope and fear are propositional attitudes.⁶ We can capture that claim as follows: there are relations R_1 and R_2 —call them “the hope relation” and “the fear relation”—such that to hope that ϕ is to bear R_1 to the proposition that ϕ , and to fear that ϕ is to bear R_2 to the proposition that ϕ . We must distinguish the fear relation from *fearing*, since one can bear the fear relations to propositions without being fearing them. (Given the belief schema, however, there is no need to distinguish “the belief relation” from *believing*.) Having introduced this useful way of talking, we can use it to state a generalisation of the belief and assertion schemas:

Attitude Schema: The proposition that ϕ is such that to bear the V-ing relation to it is to V that ϕ .

Here V can be replaced by any propositional attitude verb. It is hard to imagine a principled view that accepts the belief and assertion schemas but regards it as impossible to interpret ‘the hope relation’, ‘the fear relation’, etc. in such a way as to make the instances of the attitude schema true. However, we will not need to rely in any essential way on the general attitude schema in what follows.

Are the above schemas as unproblematic as they seem? One reason for worrying is ontological: a certain kind of nominalist will maintain that there are no propositions at all, since there are no abstract entities at all, and thus hold, for example, that the Earth is round although it is not the case that the proposition that the Earth is round is true (since there is no such thing as the proposition that the Earth is round). But philosophers who find views of this sort attractive almost always go on to introduce some piece of ideology which would allow them to make a distinction, among those sentences which they refuse officially to endorse because of their mistaken ontological presuppositions, between “good” sentences (like ‘The proposition that the Earth is round is true’) and “bad” ones (like ‘The proposition that the Earth is trapezoidal is true’). Moreover, there is reason for philosophers of this sort to take the status of ‘goodness’ to be preserved by logically valid arguments. Thus, it may be possible to set nominalistic scruples aside in order to get at the underlying issue about time, simply by explaining that our uses of sentences involving quantification over propositions is to be understood as merely aiming at the relevant form of goodness.

Proposition-talk is useful in branches of metaphysics besides the metaphysics of propositions because it allows us to achieve certain express-

⁶See King 2002, ***versus Merricks***.

ive goals. It is instructive to consider another form of language which would achieve the same goals. In logic, *second-order* languages contain variables which function syntactically like predicate letters—including variables which function syntactically like sentence letters (0-ary predicate letters). These languages thus allow for *quantification into sentence position*. We could imagine supplementing standard English with sentential variables, and the quantifiers that bind them. The grammatical sentences in the new language would include sentences like the following:

- (1) a. $\exists P(\text{necessarily, } P \text{ if and only if the Earth is round}).$
- b. $\exists P(P \text{ and John believes that } P).$
- c. $\exists P(P \text{ and it is not always the case that } P).$

There is a natural “translation” from the sentences in the new language to sentences in the old language, which involves replacing the sentential variables by ordinary objectual variables; replacing each quantifier binding a sentential variable with an ordinary quantifier restricted to propositions; adding ‘is true’ to each variable that is not the argument of some attitude verb; and replacing expressions like ‘hopes that x ’ with ‘bears the hope relation to x ’. Those who hold that such linguistic innovations depend for their intelligibility on some translation back into the original language will have no choice but to rely on some translation along these lines in making sense of quantification into sentence position. However, there is a more tolerant attitude, which allows that some philosophically interesting languages can be learnt ‘by immersion’.⁷ If you take this liberal attitude, and you think you can understand quantification into sentence position, you might find it dialectically helpful to replace claims expressed using quantification over propositions with sentences involving quantification into sentence position whose translations are logically equivalent to those claims. For example, you can substitute (1c) for the propositional temporalist’s claim ‘Some propositions are true but not always true’. If you are tempted to take a different attitude towards the resulting claim from the one you took toward the original claim, that is at least a sign that you should watch out. Perhaps something non-standard about your theory of propositions is preventing your views about propositions from fitting together with your views about the metaphysics of time in the way that would make the former a good way of getting at the latter.

* * *

⁷See, e.g., Williamson 2003, p. 459.

Having established the required background, let us return to T1 and M1. Propositional temporalists hold that just as a person can be tired sometimes without being tired all the time, a proposition can be true sometimes without being true all the time. (Indeed, typical propositional temporalists will think that whenever a person is tired sometimes without being tired all the time, the proposition that that person is tired *is* one of the propositions that is true sometimes without being true all the time.) Propositional eternalists, by contrast, hold that a proposition could no more be temporarily true than a number could be temporarily prime. Similarly, propositional contingentists hold that there are true propositions which could have been false, while propositional necessitarians hold that a proposition could no more be contingently true than a number could be contingently prime.

(The obvious way to express propositional eternalism [necessitism] is to say that each true proposition is always [necessarily] true, just as each prime number is always [necessarily] prime. Temporal [modal] modifiers are thus *inert* when applied to 'prime' and 'true'. However, some people who should clearly count as propositional eternalists and necessitarians might resist these formulations. To get a sense of the worry, consider spatial modifiers like 'somewhere' and 'everywhere'. While one could hold that spatial modifiers apply only inertly to predications of 'prime', so that 'Each prime number is prime everywhere' is true, it is not obvious that this is the right view. 'The number two is prime under my chair' is a very strange sentence; perhaps its strangeness is best explained by denying that it is true. (Compare 'The number two is happily prime' and 'The number two is prime with a knife'.) If we decide to say this, and are prepared follow the B-theorists' guiding analogy between time [modality] and space where it leads, we might be led to say regard 'The number two was prime last Tuesday', and perhaps even 'The number two would have been prime if I had missed the bus this morning', as untrue for the same reason.⁸ And if we get this far and are propositional eternalists [necessitarians], we will presumably want to treat 'true' in the same way as 'prime', leaving us with a view on which sentences like 'Some proposition was true last week' ['Some proposition would have been true if I had missed the bus'] are never true.

⁸Burgess, 'Numbers and Ideas' endorses this thought: 'Now creating a language involves creating certain rules for its use. Among these is, I believe, a rule to the effect that tense and date are not to be applied to mathematical existence assertions. One can say "There exist infinitely many prime numbers," but to ask "How many of them already existed in 1000 BCE, or during the Cenozoic Era?" is to commit a kind of grammatical solecism.'

As it happens, I think that this way of developing propositional eternalism and necessitism is a mistake: the strangeness of the relevant mathematical sentences is better explained by the claim that the relevant temporal and modal modifiers are vacuous. However, it is important not to allow the relatively shallow dispute over the proper treatment of these sentences to obscure the underlying issue.)

How is propositional necessitarianism a coherent position? The following argument for propositional contingentism seems watertight: Obama is president. But it is not necessary that Obama is president. The proposition that Obama is president is such that, necessarily, it is true iff Obama is president. Therefore, the proposition that Obama is president is true, but not necessarily true.

The key premise in the foregoing argument was an instance of the following

Modal truth schema: The proposition that ϕ is such that necessarily: it is true iff ϕ .

A propositional necessitarian could resist the argument by simply giving up on this schema. But given that we are taking the Truth Schema for granted, this looks very unpromising. If we have any understanding at all of claims of the form 'For it to be the case that ... is for it to be the case that ...', we know that they entail the corresponding necessitated biconditionals. If for a certain glass to contain water is for it to contain H₂O molecules, it is necessary that it contains water iff it contains H₂O molecules. This much should be uncontroversial: the live debate is whether there is any interpretation of necessity that validates the reverse inference from 'Necessarily P iff Q' to 'For it to be the case that P is for it to be the case that Q'. But given this, each instance of the truth schema commits us to the corresponding instance of the modal truth schema, and the argument against the propositional necessitarian stands.

There is a parallel argument for propositional temporalism. Obama is president, but he will not always be president. But the proposition that Obama is president is such that it will always be true iff Obama is president. Therefore, the proposition that Obama is president is true, but will not always be true. The key premise in this argument is an instance of the following:

Temporal truth schema: The proposition that ϕ is such that always: it is true iff ϕ

And again, given that one accepts the truth schema, a flat rejection of the temporal truth schema looks problematic. Doesn't 'For it to be the case that P is for it to be the case that Q' entail 'Always, P iff Q'? Even if one doubts the validity of the inference from 'Necessarily P' to 'Always P' (perhaps on the grounds that it is necessarily but not always true that it is raining if and only if it is raining now), these doubts do not extend to claims of the 'For it to be... ' form. (This was my motivation for formulating the truth and attitude schemas using such claims rather than necessitated biconditionals). The claim that to be water is to be H₂O manifestly rules out the suggestion that there will be water that is not H₂O in the future, or that there was any such water in the past.

Propositional temporalists and contingentists should be wary of relying on these arguments. For there is an exactly parallel argument for the (much less popular) claim that propositions can be true in some places and false in others. This argument would appeal to the following

Spatial truth schema: The proposition that ϕ is such that *everywhere*, it is true iff ϕ

Using this, we can argue from the premise that Obama is popular in California and not popular in Arkansas to the conclusion that there is a proposition (namely, the proposition that Obama is popular) which is true in California and not true in Arkansas. Few would be willing to follow this argument where it leads. But the reasons for thinking that the truth schema entails the modal and temporal truth schemas seem to carry over to the spatial case.

Perhaps, after all, propositional eternalists and necessitarians just have to deny the inference from the truth schema to the temporal and modal truth schemas. But there is a more subtle, and I think more satisfying, response which does not require rejection of any of the schemas. As we have observed, propositional eternalists [necessitarians] must posit widespread context-sensitivity in ordinary sentences. Moreover, they have reason to think that the relevant kind of context-sensitivity has an important feature in common with the kind of context-sensitivity introduced by pronouns like 'he' and 'she', a feature I will call *bindability*. By modeling their account of the relevant kind of context-sensitivity on the context-sensitivity of pronouns, propositional eternalists and necessitarians can agree that the instances of the temporal and modal truth schemas are true on all uniform interpretations.

Here is what I mean by 'bindability'. 'He is tall' is obviously a context-sensitive sentence. There is no one proposition that it must always be used

to assert, since we can use it to assert that any (salient, male) person is tall. But it is also context-sensitive in another way. As well as contributing propositions to the effect that this or that person is tall, it can also make a quite different kind of semantic contribution when it is embedded inside certain larger sentences. In 'Every farmer thinks that he is tall', for example, it is natural to interpret the pronoun 'he' as *bound* by 'every farmer'. When we are interpreting the sentence in this way, it is misguided to ask which proposition is expressed by the embedded occurrence of 'He is tall': the role played by the embedded sentence is not that of contributing any single proposition. From the propositional eternalist's point of view, there looks to be a similar contrast between unembedded uses of 'Susan is happy' and the interpretation it naturally bears when embedded under temporal modifiers, e.g. in 'Whenever there is a party, Susan is happy'. Given propositional eternalism, it seems odd to think of this embedded occurrence of 'Susan is happy' as contributing, or bearing any semantically important relation to, some particular proposition, such as the proposition that Susan is happy at the particular time *t* at which the utterance is taking place. Associating such a proposition with the embedded occurrence does not helpfully characterise its role in determining the proposition expressed by the whole sentence.⁹

The dominant view in the semantics of tense takes the analogy between 'Susan is happy' and 'He is tall' very seriously. On this view, some syntactic constituent of 'Susan is happy' (perhaps the present tense morpheme on 'is', or some unvoiced constituent) has the semantic role of a *variable* ranging over times. Such variables are relevantly similar to the pronoun 'he', in being able to make two quite different kinds of semantic contribution: they can be bound by higher quantifiers (like 'always' and 'whenever there is a party'), but they can also be left unbound, in which case they will refer to some particular time in the same way that an unbound occurrence of 'he' refers to some particular person. But whether or not they embrace this particular style of syntax and semantics, propositional eternalists have good reason to think that somehow or other, embedded occurrences of tensed sentences admit distinctive interpretations that do not correspond to any interpretations these sentences can bear when unembedded.

This distinction opens up a new line of response to the arguments from the modal, temporal and spatial truth schemas to propositional contingentism, temporalism, and 'spatialism'. Recall that in accepting any of schema of this sort, we are committed to the truth of its instances *only when repeated*

⁹***Cite Salmon 1986.

context-sensitive constituents are interpreted uniformly. Consider the instance of the temporal truth schema that was appealed to in the argument for temporalism:

- (2) The proposition that Obama is president is such that always, it is true iff Obama is president.

Here the second occurrence of ‘Obama is president’ cries out to be interpreted as bound by ‘always’, since if we do not interpret it like this, we will have to regard ‘always’ as vacuous. And even if there is a semantically legitimate interpretation of (2) on which ‘always’ is vacuous, it is not one we find it easy to access, for reasons that are easily understood.¹⁰ Compare ‘Every farmer thinks that farming comes naturally to him’ with ‘Every farmer is such that farming comes naturally to him’. An unbound reading of ‘him’ as referring to some salient male person is much easier to hear for the former, since in the case of the latter, such a reading would make ‘Every farmer is such that’ completely vacuous. But if we interpret the second occurrence of ‘Obama is president’ in (2) as bound, we will not be interpreting the sentence uniformly. For since the first occurrence of ‘Obama is president’ in (2) is not in the scope of any temporal modifier, it must be understood as contributing some particular proposition, such as the proposition that Obama is president at *t* (where *t* is the time of utterance). The upshot is that while (2) comes out false on its *natural* interpretation, this is no objection to the temporal truth schema, since the interpretation in question is a non-uniform one.

For an analogy, consider the following obviously correct schema (which follows from the belief schema and the truth schema):

- (3) If ϕ , then everyone who believes that ϕ believes something true

It is no objection to (3) that the following instance admits of a false interpretation where the first ‘he’ is used to demonstrate some particular person, and the second is bound by ‘everyone’:

- (4) If he is tall, then everyone who believes that he is tall believes something true.

By the propositional eternalist’s lights, apparent counterexamples to the temporal truth schema like (2) can be dismissed in the same way as (4). (The

^{10***}ADD: discussion of Percus 2000 on the question whether such a reading is even possible.

only relevant difference is that whereas uniform readings of (4) where both 'he's refer to some particular person are easy to access, uniform readings of (2) are either impossible or at least very hard to access because of our bias against interpretations on which certain constituents turn out to be vacuous.) The same move can be used to defeat the inferences from the modal and spatial truth schemas to contingentism and "spatialism".¹¹

The literature on propositional temporalism and eternalism is haunted by the apprehension that the whole issue might turn out to be in some sense a verbal one. For example, Kaplan, in introducing a view which seems to countenance propositions whose truth values change over time, uses scare quotes around 'proposition', in order, as he puts it, to 'reflect my feeling that this is not the traditional notion of a proposition' (Kaplan 1989, n. 28); he contemplates a view which seems to involve positing propositions whose truth value varies over space, and describes the question whether to adopt such a view as 'largely a matter of linguistic engineering' (p. 504). Remarks like this suggests that there might be at least two legitimate possible uses or senses of 'proposition', and perhaps also of 'true', such that T1 receives different answers depending on how we interpret it.

But if we lay down the truth schema as a non-negotiable constraint on the interpretation of 'proposition' and 'true', it is hard to see how this conciliatory view could be maintained. Suppose there are two available interpretations of

¹¹In formal semantics, the usual way to register facts about the binding of variables uses a system of numerical indices: a variable is bound when it is in the scope of a quantifier carrying the same index. In such systems, there is normally nothing to stop indices from being "recycled". For example, if we posit a variable ranging over times in the syntax of 'Obama is president', there is nothing to stop us from attaching the same index number to the unbound occurrence of that variable in the first occurrence of 'Obama is president' in (2), to the quantifier in 'always', and to the bound occurrence of the variable in the second occurrence of 'Obama is president'. *Prima facie*, this seems to go against my claim that a reading on which the first occurrence is unbound and the second occurrence is bound involves a 'non-uniform' resolution of context-sensitivity. For in a system with indices, it is tempting to cash out the relevant notion of uniformity in terms of the sharing of indices.

However, while the recycling of indices is a feature of standard logical notation, it doesn't seem to correspond to any genuine phenomenon in natural language. There is nothing that would mark the difference between utterances of (2) in which the earlier unbound variable shares an index with the later bound variable, and utterances in which it does not. It seems safe, then, to dismiss this kind of arbitrary index-sharing as an artefact stemming from the use of numerical indices, a practice towards which it is in any case reasonable to adopt a somewhat instrumentalistic attitude.

‘proposition’ and ‘true’, both of which obey the truth schema, such that some propositions₁ are temporarily true₁, but no propositions₂ are temporarily true₁. By the two versions of the truth schema, we have the following:

- (5) The proposition₁ that Obama is president is such that for it to be true₁ is for Obama to be president.
- (6) The proposition₂ that Obama is president is such that for it to be true₂ is for Obama to be president.

These jointly entail (7):¹²

- (7) The proposition₁ that Obama is president and the proposition₂ that Obama is president are such that for the former to be true₁ is for the latter to be true₂.

Given that ‘For it to be the case that *P* is for it to be the case that *Q*’ entails ‘Always, *P* iff *Q*’, (7) yields

- (8) The proposition₁ that Obama is president and the proposition₂ that Obama is president are such that always, the former is true₁ iff the latter is true₂.

thereby ruling out the supposed contrast between propositions₁ and propositions₂.

One could resist this argument by appealing to the conception of ‘always’ as a binder. Perhaps the truth of (7) only requires the truth of (8) when we (unnaturally) read the ‘always’ in the latter as vacuous, so as to preserve an unbound interpretation for all the material in its scope. This leaves open the possibility that (8) has a more natural reading on which it is false. But the view of ‘always’ as a binder—or at least, the version of that view on which the only way to achieve a uniform interpretation of a sentence or argument in which some sentence occurs sometimes in the scope of ‘always’ and sometimes outside it is to read ‘always’ as vacuous—is not neutral as between temporalists and eternalists: it effectively presupposes eternalism.

¹²I am assuming here that ‘For it to be the case that *P* is for it to be the case that *Q*’ has the logic of an equivalence relation. This is controversial. Some think of these kinds of claims as entailing something about the ‘priority’ of one side over the other; they will thus deny that ‘For the glass to contain water is for it to contain H₂O’ entails ‘For the glass to contain H₂O is for it to contain water’. But even these philosophers should have no trouble introducing a subsidiary reading of the ‘For it to be the case. . .’ claims on which this inference goes through; they should interpret my claims as involving this notion.

If there were temporarily true propositions, we could use a sentence to express the very same such proposition both inside and outside ‘always’, without ‘always’ being vacuous. Since it is of the essence of the conciliatory view that it is neutral between temporalism and eternalism (regarding each as true on some reading of ‘proposition’ and ‘true’), it is unsatisfactory if the only way to defend it from the foregoing objection involves appealing to a conception of the demands of uniformity that presupposes eternalism.

1.3 Other A-theoretic themes

The dispute between “A-theorists” and “B-theorists” about time has been going on for over a century. In that time, a multitude of definitions have been given of the two positions, not obviously equivalent to one another or to any combinations of answers to my questions T1 and T2. It is not my purpose in this book to provide an overview of this body of literature. I hope it is fairly plain from the collection of quotes on p.*** that my questions are at least *relevant* to the historically important debate, that a certain kind of paradigmatic A-theorist would give a positive answer to both, and that a certain kind of paradigmatic B-theorist would give a negative answer to both. I am not going to argue that my questions exhaust what people have been disagreeing about, or that they are the only intelligible questions in the vicinity. However, I do have a reason for spending a bit of time considering whether there is any coherent and defensible further commitment, not entailed by the combination of T1 and T2, that might reasonably be regarded as integral to the A-theoretic vision. The reason is that I would like to persuade people who are already aligned with the A-theoretic camp to take seriously the distinctive theory I will be developing in the later parts of this book. This theory includes what I like to think of as the central claims of the A-theory, namely propositional temporalism and temporal elitism; but I predict that some A-theorists will nevertheless feel that it isn’t giving them what they really wanted. So, let’s consider some of the ideas that often come up in explanations of the A-theory/B-theory debate whose relation to T1 and T2 is not obvious.

(i) *Flow*

It is hard to make sense of the thought that *possible worlds* undergo “flow” or “passage”. Certainly the combination of propositional contingentism and

modal elitism does not commit us to any such mysterious doctrine! This naturally suggests that the common A-theoretic slogan that *time* “flows” or “passes” has some content that goes beyond what is captured by the combination of propositional temporalism and temporal elitism.

Fine (2005c) thinks that this is a serious problem for that combination, or at least for its motivation. He expresses the concern as follows:

The standard realist [roughly: the propositional temporalist temporal elitist] faces a general difficulty. For suppose we ask: given a complete tenseless description of reality, then what does he need to add to the description to render it complete by his own lights? The answer is that he need add nothing beyond the fact that a given time *t* is present, since everything else of tense-theoretic interest will follow from this fact and the tenseless facts. But then how could this solitary ‘dynamic’ fact, in addition to the static facts that the antirealist is willing to accept, be sufficient to account for the passage of time? [The realist’s] conception of temporal reality. . . is as static or block-like as the anti-realist’s, the only difference lying in the fact that his block has a privileged center. Even if presentness is allowed to shed its light upon the world, there is nothing in his metaphysics to prevent that light being ‘frozen’ on a particular moment of time. (Fine 2005c, p. ***)¹³

To this, temporalist-elitists will reply that it is part of their view that the light of presentness is *not* ‘frozen on a particular moment of time’, since their view entails that other, future times *will* be present, and that other, past times *have been* present. What more could it conceivably mean for the light not be frozen? Fine anticipates this response, but finds it unconvincing on the following grounds:

¹³Note that Fine’s ‘anti-realists’ are not standard temporal B-theorists. While they are temporal egalitarians—as Fine puts it, they hold that ‘no time is privileged’—they accept propositional temporalism, which Fine regards as obviously true. ?? discussed Fine’s strategy for making sense of this combination.

I am not sure what Fine means by a ‘complete tenseless description of reality’. If ‘tenseless’ just means ‘necessarily either eternally true or eternally false’, the assumption that the proposition that according to the realist is a complete tenseless description of reality is one that ‘the anti-realist is willing to accept’ is highly problematic. The realist might very well think that temporal elitism itself is not only true, but necessarily and eternally true, while the anti-realist of course thinks that temporal elitism is false. Fine may intend ‘tenseless’ in some more demanding sense, but I don’t know what this sense might be, or why ‘everything else of tense-theoretic interest’ is supposed to follow from the facts that are tenseless in the *demanding* sense and the truth about which time is present.

However, the future presentness of $t+$ amounts to no more than t being present and $t+$ being later than t and, similarly, the past presentness of $t-$ amounts to no more than t being present and $t-$ being earlier than t . But then how can the passage of time be seen to rest on the fact that a given time is present and that various other times are either earlier or later than that time?

It is certainly true, given the standard way of thinking about instants (which ?? will attempt to lay out systematically), that the following two claims are necessarily equivalent:

- (9) a. $t+$ will be present
- b. $t+$ is after the instant that is present

(It is unclear whether this necessary equivalence justifies Fine's 'amounts to no more than', which suggests that (9b) is in some sense prior to (9a); but let us not dwell on this.) Fine seems to find it mysterious how there *could* be any such necessary equivalence as this. At least, he seems to find it mysterious how (9a) and (9b) could be equivalent if 'will be' is given its ordinary meaning, the one that makes (9a) inconsistent with the idea that presentness is 'frozen' on an instant distinct from $t+$. Here are two instants, t and $t+$, which stand in a certain relation, *afterness*. How (the thought runs) could this relation, together with the fact that t is present, necessitate anything about where presentness will go next, if it goes anywhere? If presentness in fact is going to flow along the timeline at a rate of one second per second, how could that be a strictly necessary truth as opposed to a mere law of nature?

I'm not sure if this is the thought behind Fine's rhetorical question. But thoughts along these lines have, I think, been quite influential in the debate about the A-theory. Indeed, some A-theorists seem to have implicitly accepted that the equivalence between (9a) and (9b) is *not* metaphysically necessary, and engaged in strange and palpably wrongheaded metaphysical speculations (for example, speculations about the existence of multiple temporal dimensions) in trying to allow for the possibility that things are otherwise. And the thought that A-theorists are forced to go down this route seems to be the basis of certain famous objections to the A-theory (e.g. Smart 1966). But it is hard to see where the mystery is supposed to lie. Why shouldn't (9a) and (9b) be metaphysically necessarily equivalent?

One way to make a putative metaphysical necessity seem unmysterious is to show how it follows logically from a 'real definition' or 'metaphysical

analysis'—some claim of the form 'To be F is to be G', or 'For it to be the case that P is for it to be the case that Q'. In the case of (9a) and (9b), there are two well-known ways of doing this. We could, first, analyse 'after' in terms of tense operators, as follows:

- (10) For $t+$ to be after t is for it to be the case that always, if t is present, $t+$ *will* be present (**cite Prior**).

Alternatively, we could analyse tense operators in terms of 'at' and 'after':

- (11) For it to be the case that it will be that ϕ is for it to be the case that at some instant that is after the present instant, ϕ .

If we endorsed (10) or (11), there would be nothing mysterious about the necessary equivalence of (9a) and (9b). And what is supposed to be the principled objection to real definitions like (10) and (11)? Some putative real definitions are unbelievable on the grounds that they relate facts that are intuitively metaphysically distinct. But (10) and (11) are not at all like this. It is obvious that there is a very intimate relation between facts about how things are before and after the present and facts about how things will be and have been; I see nothing strange or repugnant about the suggestion that these two kinds of facts come to the same thing.

Thus, I can't see any intelligible further question about "flow" or "passage" that is not settled by the combination of propositional temporalism, temporal elitism, and the necessary equivalence of (9a) and (9b). Moreover, that necessary equivalence strikes me as something we have every reason to accept if we are going to go in for talk of instants at all.

Chapter 2

Arguments about A- and B-theories

2.1 The psychological objection to the B-theories

On the best-known versions of propositional eternalism, the propositions we use ordinary present-tensed sentences to assert are *de re* propositions about the times at which we utter them. For example, someone who is uttering ‘Harvey is asleep’ at *t* will thereby be asserting the proposition *that Harvey is asleep at t*. This is sometimes called the ‘date theory’ of tensed sentences.

There is a famous objection to the date theory, according to which it misses something crucial about the psychological states we use these sentences to express and to induce in others. It is easiest to state the objection if we assume that expressions like ‘2 am on the 16th of June, 2011’ are directly referential, so that

- (1) Harvey is asleep at 2 am on the 16th of June, 2011

expresses the proposition, concerning that particular instant, that Harvey is asleep at *it*. Given this assumption, the worry can be raised by considering pairs of attitude ascriptions like

- (2) John believes that Harvey is asleep
- (3) John believes that Harvey is asleep at 2 am on the 16th of June, 2011

It seems that one could speak truly in asserting one of these sentences and falsely in asserting the other even if the time of one’s speech is 2 am on the 16th of June, 2011. For John might fail to believe that it *is* 2 am on the 16th of June, 2011—he might be wrong about the time. But if the date theory is true, then as used at that time, (2) and (3) will express the same proposition.

The assumption that (1) expresses a *de re* proposition about a time is not really essential to this problem. For suppose John has lost track of the time

but knows exactly when Harvey (an amazingly regular sleeper) is asleep; then I could, it seems, speak falsely in uttering (2) while speaking truly in uttering

- (4) Harvey is asleep; and for every time at which Harvey is asleep, John believes that Harvey is asleep at that time.

By contrast, propositional temporalists have no trouble explaining how (2) could be false even while (3) and (4) are true. By their lights, there is no reason why John's knowledge of eternal propositions should not be combined with ignorance of the non-eternal proposition *that Harvey is asleep*.

One could attempt to run a parallel objection to the modal analogue of the date theory—the view that at any given world w , utterances in English of a given sentence ' P ' (not otherwise context-sensitive) express the proposition that at w , P . This objection would turn on cases where, at w , certain people know that at w , P without (intuitively) knowing that P . However, it is harder to get this argument off the ground in the modal case than it was in the temporal case, since it is far from clear how to think about the achievement involved in believing a *de re* proposition about a particular world. How are the people in the allegedly possible scenario supposed to *pick out* w in order to think that at w , P , given that they do not simply think of it as 'the actual world'? Some (Soames 1998, Williamson 2000b) have suggested that entertaining a *de re* proposition about a world (without just picking it out as 'the actual world') requires enormous cognitive resources, sufficient to formulate a qualitative description of how things are at the relevant world that distinguishes it from all other worlds. But this may be too demanding: it is arguable that we could in some cases entertain such propositions using descriptions like 'the closest world at which such-and-such is the case'. Indeed, there are some views on which *de re* thought about worlds comes very easily indeed. For example, if there are a million cats at w , and one thinks at w' that there are a billion cats at w' , some might think this enough for it to be true that at w' , one thinks that there are more cats at w' than at w . (Cf. 'If there had been a billion cats, I would have known that there were more cats than there actually are'.) Views of this sort might make the modal analogue of the psychological objection to the B-theory quite easy to run.

The psychological objection is not just that the target theory lacks an adequate account of certain sentences involving attitude verbs. The basic idea behind the objection (in the temporal case) can also be used to argue against the claim that an utterance of 'Harvey is asleep' at 2am on the 16th of

June, 2011 would express the same proposition as an utterance of (1). For if I am talking to someone who knows all about Harvey's regular sleep habits but has lost track of the time, 'Harvey is asleep' could be a helpful thing to utter. But according to the date theory, in uttering this I would be asserting something that my hearer already knows, which doesn't sound like a helpful thing to do at all.

Psychological considerations of this kind have led many philosophers to endorse propositional temporalism.¹ But they should not have been so easily convinced. While the foregoing arguments may or may not be good arguments against the date theory, they are certainly not good arguments against propositional eternalism, or for that matter against propositional necessitism. To see this, consider the parallel between the argument against the date theory and the following arguments:

- (i) 'The Babylonians believe that Hesperus is visible in the evening' and 'The Babylonians believe that Phosphorus is visible in the evening' can differ in truth value. So 'Hesperus is visible in the evening' and 'Phosphorus is visible in the evening' cannot both semantically express the proposition that x is visible in the evening, for any x . Such an account would wrongly entail that the sentences can be substituted *salva veritate* in attitude reports. It would also leave us with no way to explain why 'Phosphorus is visible in the evening' might be a helpful thing to assert, even when we are talking to someone who already knows that Hesperus is visible in the evening.
- (ii) Utterances of 'I believe I am in Oxford' and 'I believe Dorr is in Oxford' could differ in truth value, even if Dorr is the speaker in both cases. Thus, it cannot be that what 'I am in Oxford' expresses when uttered by Dorr is the proposition that Dorr is in Oxford. Such an account would wrongly entail that the sentences can be substituted *salva veritate* in attitude reports (when the speaker is Dorr); it would also leave us with no way to explain why 'I am in Oxford' might be a helpful thing to assert, even when we are talking to someone who already knows that Dorr is in Oxford (but does not know that the person speaking is Dorr).²

¹*Lots of citations here!*

²Many philosophers have noted that the fact that the psychological phenomena that drive the argument seem to work in the same way for first-person thinking as for present-tense thinking makes it hard to see how these phenomena could possibly motivate temporalism.

There have been two broad reactions to the Hesperus-Phosphorus argument. *Russellians* reject the argument. According to them, there *is* a planet x (namely, Venus), such that ‘Hesperus is bright’ and ‘Phosphorus is bright’ both semantically express the proposition that x is bright. The best-known form of the Russellian view (that of Salmon and Soames ***cite***) combines this with the claim that the single proposition expressed by both ‘The Babylonians believe that Hesperus is visible in the evening’ and ‘The Babylonians believe that Phosphorus is visible in the evening’ is *true*, relegating any considerations that might make the latter sentence a bad thing to assert to the realm of pragmatics. According to these Russellians, it is often a perfectly good idea to tell people things that they already know. For our asserting these already-known propositions can have other desirable effects: it can put our hearers to in a position to draw inferences they were not previously in a position to draw, or allow them to bring their knowledge to bear on their actions in a way that they were previously unable to do. This is often explained by appealing to the idea that propositions can be entertained under different ‘guises’. Hearing ‘Phosphorus is visible in the evening’ will enable people to know the proposition semantically expressed *under a new guise*, and this is often an achievement of great significance. Another form of Russellianism involves positing a distinctive form of context-sensitivity in sentences involving attitude verbs. The idea is that while ‘Hesperus’ and ‘Phosphorus’ have the same semantic value, the choice to use one or the other in a belief report can make certain resolutions of its context-sensitivity more natural than others. On this account, there is one reading of ‘know’ on which those who hear ‘Phosphorus is visible in the evening’ ‘come to know something new’, while there is a different reading on which they do not. Contextualist Russellians may also find a role in their theory for the idea of ‘guises’, now as part of their account of the range of interpretations which attitude reports can bear.

Those who are satisfied with the Russellian view of ‘Hesperus’ and ‘Phosphorus’ (and the analogous views about ‘groundhog’ and ‘woodchuck’, ‘attorney’ and ‘lawyer’, etc.) should, I think, be unmoved by the psychological objection to claim that ‘I am in Oxford’ expresses the same proposition as ‘Dorr is in Oxford’, and by the psychological objection to the date theory. They can tell their story in terms of “guises”. Someone who has lost track of the time but knows general truths concerning Harvey’s sleep habits will know the proposition that Harvey is asleep at t under one guise—a guise as-

sociated with expressions like ‘2am on the 16th of June, 2011’—while failing to know this proposition under another, very important, guise (“the present-tense guise”). Sentences ‘Fred knows that Harvey is asleep’ have a certain association with the present-tense guise: perhaps they pragmatically convey that the relevant proposition is known under its present-tense guise, or perhaps they naturally favour a resolution of the context-sensitivity of ‘knows’ under one has to employ the present-tense guise to count as ‘knowing’ *de re* propositions about a time, at that time.³

The Fregean response to the ‘Hesperus’-‘Phosphorus’ case, meanwhile, goes along with a different picture: where Russellians see a single proposition and many guises, Fregeans see many propositions. For Fregeans there are, for example, a vast variety of propositions that predicate the property *being visible in the evening* of Venus. While these propositions necessarily all have the same truth-value, they differ in what it takes to bear propositional attitudes towards them. In hearing ‘Phosphorus is visible in the evening’, I will indeed come to know a new proposition, albeit one necessarily equivalent to a proposition I already knew. This view of the realm of propositions opens up new possibilities for context-sensitivity in semantics, and new distinctions between varieties of context-sensitivity. Even if we lay down that sentences involving ‘I’ always express propositions about the person who is uttering them, we are free to develop different accounts of what it takes to bear attitudes towards the propositions in question. ‘Dorr is in Oxford’ could express a proposition such that it is sufficient for believing it that one believe that the person named ‘Dorr’ is in Oxford, while Dorr is in fact the person named ‘Dorr’. Meanwhile, in the context of a typical utterance by Dorr, ‘I am in Oxford’ could express a proposition that is harder to believe—for example, perhaps it is part of the nature of this particular proposition that *Dorr* gets to bear the belief relation to it only when he is in a position

³The Russellian defence of the date theory *could* be combined with a picture of propositions according to which propositions that necessarily have the same truth value are always identical. But this view faces the problem of logical omniscience (anyone who believes that all dogs are dogs believes all logical truths); explaining away the apparent counterexamples to logical omniscience pragmatically, or relying on some story about context-shifting, seem like desperate options. For these reasons, Russellians have generally preferred to work with a picture of propositions as structured (Salmon 1986, Soames 1987). Propositional necessitarians who favour the Russellian response to the psychological objection had better go for structured propositions of some sort. Otherwise they will be left with a view according to which there is just one true proposition and one false proposition. Defending this view by appeal to pragmatic effects or context-shifting seems hopeless.

to say to himself, 'I am in Oxford'. (Though presumably this is not required for *others* to believe it.) This explains how typical utterances of 'I believe that Dorr is in Oxford' and 'I believe that I am in Oxford' could differ in truth value. However the details go, a Fregean picture of propositions will make it easy to assign intuitive truth values to attitude reports while keeping an account of 'I' as context-sensitive. And this point carries over to tense. Even if 'Harvey is asleep' and 'Harvey is asleep at 2 am on the 16th of June, 2011' would typically express necessarily equivalent and structurally isomorphic propositions, if both were uttered at 2 am on the 16th of June, 2011, the propositions in question might differ in the conditions required for bearing attitudes towards them. In particular, it might be that believing *at that instant* the proposition expressed by 'Harvey is asleep' requires being in a position to say to yourself 'Harvey is asleep *now*'. There is no threat here to the thought that the proposition expressed is eternally true or false. (An analogous response can be made to the argument against propositional necessitarianism.)

I said earlier that the psychological argument has led many philosophers to endorse propositional temporalism. However, it is not always clear whether the philosophers who say things that sound like endorsements of propositional temporalism in response to the psychological argument really are endorsing propositional temporalism in our sense, because it is not clear whether they are using 'proposition' in a way that conforms to the role outlined in section 1.2. Consider the following passage from Sider (2002):

Let us understand *propositions* atemporally, as being true or false *simpliciter*, not relative to time. Prior succeeds in showing that psychological attitudes are not simply relations to propositions, thus understood. A more appropriate object of the attitudes would be what one might call 'temporal propositions': functions from times to (atemporal) propositions.

Sider is here drawing on idea from Lewis (1979), who similarly argues that the "objects of the attitudes" are 'centred propositions', something like functions from $\langle \text{world, time, individual} \rangle$ triples to truth values. However, the central claims in Lewis 1979 do not require the view that such functions are ever designated by terms of the form 'the proposition that ϕ ', or that sentences of the form '*S* believes that ϕ ' ever characterise their subject as standing in psychological relations (such as the relation Lewis calls 'self-ascribing') to particular such functions. To see why this is not a requirement

of Lewis's theory, notice that a semantic theory that took 'believes' to express the self-ascribing relation, and 'I am hungry' to semantically express the Lewisian 'centred proposition' that maps $\langle w, t, x \rangle$ to Truth iff x is hungry at t at w would be a complete non-starter. For it is clearly false that for any x , x believes that I am hungry iff x self-ascribes hungriness. Lewis of course is well aware of this: he suggests a more complicated view on which what it takes for x to believe that I am hungry is for there to be a relation R , meeting certain conditions, such that (i) x bears R to me, and (ii) x self-ascribes *bearing R to something hungry*. It is consistent with Lewis's view that attitude ascriptions in which names and variables occur in the scope of an attitude verb are always interpreted in this "de re" way. (He does not discuss how these truth-conditions are to be generated compositionally; there are various ways in which this might be done, many of which do not involve taking 'believes' to express the self-ascribing relation.) Thus, when Lewis says that centred propositions are the "objects of the attitudes", the central idea is not that attitude verbs express relations to centred propositions; rather, it is that such relations play an certain distinctive role to play in an account of the *metaphysics* of the attitudes, an answer to the question what it *is* to believe (or desire or hope or assert or be confident) that P . And this claim is perfectly consistent with wholehearted propositional eternalism in the sense we are interested in. For example, it is consistent with a propositional eternalist view on which sentences express structured Russellian propositions, and ' x believes at t that ϕ ' is true at w iff there is a relation R between people, times, and structured propositions, meeting certain conditions, such that (i) R holds at w between x , t , and the semantic value of ϕ , and (ii) x self-ascribes the function that maps x' , t' , and w' to the True iff R holds at w' between x' , t' , and some true Russellian proposition.

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